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Computerized inhibitory control training reduces anxiety in preadolescent students

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ABSTRACT

The study examined the relationship between anxiety, inhibitory control (IC), and resting-state electroencephalography (EEG) in a critical age-range for social and emotional development (8–12-year-olds). The participants were assigned to 4 weeks of either an emotional IC training program, a neutral IC training program, or a waitlisted control, and were tested using cognitive, emotional, and EEG measures. The training was computerized and completed remotely. At baseline, IC accuracy scores were negatively related to both anxiety and depression levels ($N=42$). Additionally, increased right lateral frontal alpha asymmetry was predictive of increased anxiety/depression scores. A series of multivariate analyses of covariance and post-hoc tests were conducted to compare effects in the participants that completed the full 16 sessions of training ($N=32$). Overall the emotional and neutral training conditions showed similar improvements in IC accuracy, as well as reductions in anxiety compared to the waitlist condition. Minimal neurophysiological changes occurred from pre-to-post; however, lateral frontal asymmetry shifted leftward in the emotional training group. These findings highlight the potential of computerized IC training for mitigating negative emotional functioning in preadolescents. Future research is necessary to determine the long-term effects of IC training and whether longer training intervals facilitate persisting impacts.

KEYWORDS

Anxiety; inhibitory control; computerized inhibitory control training; resting-state eeg

Societal anxiety levels in the U.S. have risen in recent years and have particularly intensified in youth demographics (Lagoe & Atkin, 2015). According to the Anxiety and Depression Association of America (ADAA), anxiety disorders affect 31.9% of children between the ages of 13 and 18 years (38% for females and 26.1% for males). Given the predominately adverse influence of anxiety on social, psychological, and cognitive functioning; early prevention, management, and quality treatment plans are critical topics of modern research (Dadds & Roth, 2008). Traditional treatments of anxiety and depression including cognitive-behavioral therapy (CBT) and psychiatric medications are somewhat successful in alleviating adult symptoms, however they are financially exigent and have yielded some mixed results in children (Storch & McKay, 2013; Anderson et al., 2012). Therefore, discovering appropriate means for reducing childhood anxiety and depression which are affordable and accessible is paramount.

Anxiety involves strong cognitive influences, which are both explicit (obsessive thought processes and ruminations) as well as implicit (negative processing bias and reduced cognitive control) (Hallion et al., 2017; Vytal et al., 2012). Advancements in technology have made it possible to train certain cognitive abilities using child-friendly applications or games that can be easily accessed from a home computer (Nouchi et al., 2012). Devising computerized training programs which target underlying cognitive characteristics

related to anxiety is a promising method for attenuating symptoms and risk in children (Joormann & Vanderlind, 2014).

Inhibitory control functioning and anxiety

Inhibitory control (IC) is broadly defined as the ability to suppress an initial predominant response (thought or behavior) in favor of a more optimal or appropriate one (Best & Miller, 2010). IC has emerged as a key subcomponent of the broader construct of executive functioning (via factor analysis) along with working memory and mental flexibility (Miyake et al., 2000; Rose et al., 2012). Executive Functioning refers to a critical set of cognitive abilities driven by the dorsolateral prefrontal cortex that are involved in solving tasks, planning for the future, and exhibiting appropriate social behaviors. IC impairment is hypothesized to be a cognitive mediator of anxiety through both explicit and implicit mechanisms. The explicit theory of generalized anxiety disorder (Geiger et al., 2016; Moran, 2016) explains that the manifestation of anxiety occurs in part due to the inability of individuals to effectively recognize unrealistic or overly critical thinking patterns (ruminations) and suppress them.

Implicit conjectures such as the attentional control theory, imply that an increased susceptibility for anxiety will be present in individuals that are more easily distracted or

interfered by input in the environment (Derakshan & Eysenck, 2009). The antisaccade task, where one is instructed to direct eye gaze in the opposite direction of a cued stimuli has been useful for studying such theories. Adults with anxiety have shown longer latencies on antisaccade trials compared to “normal” controls, with no group differences occurring on prosaccade trials (Ansari & Derakshan, 2010). Follow-up work found that this characteristic was linked to depleted frontoparietal activity immediately preceding the stimuli (Ansari & Derakshan, 2011). In a similar investigation using the Stroop task, Price and Mohlman (2007) found impaired task performance (accuracy) in elderly adults with GAD, but in contrast to Ansari and colleagues, found reduced reaction times in this group compared to controls. Further work from Edwards et al. (2017) found that cognitive trait anxiety was linked to both inhibitory control accuracy decreases and reaction time increases (using a Go-No-Go paradigm); in contrast, somatic trait anxiety was unrelated to these outcomes.

Anxiety-related impairments in IC may be more pronounced in tasks containing content that elicits emotional processing or valence, which are often referred to as “Hot EF” tasks. Yoon et al. (2017) assessed two groups (adults with GAD and neurotypical controls) on a reading-span task with both neutral and emotional distractors. They found that individuals with GAD scored lower on the reading span task, and effects were intensified when the words were designed to evoke emotional response (Yoon et al., 2017). The reading span task demands a high degree of inhibitory control because it is necessary to filter out content that is irrelevant or distracting to the memorization of the stimuli sequence (Geiger et al., 2016).

Thus far, limited research has been devoted to exploring the IC-anxiety association in child or adolescent samples. A meta-analysis from Lipszyc and Schachar (2010) found IC deficits (using the stop-signal task) in ADHD ($g = 0.62$) obsessive compulsive disorder ($g = 0.77$) and schizophrenia ($g = 0.69$), but not anxiety disorders. Notably, the review of anxiety studies comprised of six investigations featuring child samples and two with adult samples. Mueller et al. (2012) compared 22 preadolescents with anxiety disorder ($M = 11.97$) to 22 age-matched, non-anxious controls ($M = 12.51$) on an emotional antisaccade task. Participants with anxiety showed impaired accuracy but swifter reaction times on angry expression trials compared to controls. Significant differences did not occur for happy, fearful, or neutral expressions trials. In summary, anxiety has been connected with impaired IC accuracy (primarily in adult samples) while the results have been more mixed regarding IC reaction time.

Effectiveness of computerized executive functioning training

In recent years, various brain-training methods have been developed aiming to improve EF and IC, and strengthen corresponding PFC circuitry on the basis of neuroplasticity (Draganski & May, 2008). Research evaluating the effectiveness of such training has yielded mixed results, with some

studies reporting cognitive improvements, often depending on the length, adaptivity, and type of training (Melby-Lervåg & Hulme, 2013; Morrison & Chein, 2011; Nouchi et al., 2012). In certain investigations, the computerized training of EF has also led to reductions in mood-disorder symptoms, especially ruminations (Bomyea & Amir, 2011).

Sari et al. (2016) assessed the impact of a 3-week daily dual n-back training and found improvements on an inhibitory control measure (flanker task) as well as reductions in self-reported anxiety in young adults. Bomyea and Amir (2011) tested the efficacy of a working memory training in reducing daily ruminations in undergraduate students. In a direct comparison between WM training with high-inhibition and low-inhibition, there was a significantly greater reduction of intrusive thoughts in the group trained with high inhibitory demands. Similarly, Cohen et al. (2015) showed training related gains in state rumination using a delayed match-to-sample training task that emphasized inhibition (images projected in-between trials). Together, these results demonstrate the importance of encapsulating elements of IC in cognitive training paradigms aimed at reducing anxiety.

Still, certain studies using computerized training programs have not observed any reductions in anxiety, depression, or ruminations (Onraedt & Koster, 2014; Wanmaker et al., 2015). To optimally train the emotional aspects of cognitive control as well as brain areas involved in the top-down control of emotions (dorsolateral PFC, cortico-limbic circuit), it may be optimal to comprise tasks with emotional stimuli or distractors that evoke emotional response (De Voogd et al., 2016). Promising evidence has emerged for such training from both Schweizer and colleagues, as well as Iacoviello et al. (2014). Schweizer et al. (2013) found a reduction of trait anxiety and social anxiety resulting from an emotional dual n-back (WM) training given to participants aged 23–25 years, for 20 days in total. Iacoviello et al. (2014) administered a comparable task referred to as the emotional faces matching task (EFMT) to adults with major depressive disorder, and found that following eight sessions of training the EFMT led to a greater reduction in depressive symptoms than a control task featuring shape stimuli instead of emotional faces. In De Voogd et al. (2016), a similar emotional EF training (checkerboard task) was delivered to adolescents (11–18 years) and facilitated significant improvements in self-esteem, but no reduction of anxiety symptoms. It should be noted that training was only delivered twice weekly, over a span of four weeks. In summary, research on EF/IC training to date has focused predominately on adult samples, and has only used self-report measures to operationalize anxiety or depressive symptoms. Therefore, we extended this research to a younger sample consisting of 8–12-year-olds and included an objective outcome measure (resting-state EEG) to enable more succinct conclusions about training efficacy.

Neurophysiological profile of anxiety/depression

Resting-state brain activity measurements have been useful for revealing neural characteristics related to anxiety and

depressive disorders. EEG is advantageous for developmental research because as a noninvasive technique it does not require any x-rays, radiations, or injections and is becoming increasingly common in hospital and mental health care settings. In past studies, EEG has also been utilized as a tool for assessing the effectiveness of various interventions by comparing activity profiles before and after treatment (Gollan et al., 2014; Moscovitch et al., 2011; Oathes et al., 2010).

In children, adolescents, and adults, depressive and anxiety disorders are frequently associated with a pattern of right frontal alpha asymmetry (Davidson et al., 2000; Henriques & Davidson, 2000; Jones & Field, 1999), indicating greater left than right power but greater right than left activation, due to the inverse relation of alpha power and brain activity. Coan and Allen (2004) suggested that EEG data could serve as a diathesis-stress framework, in which an atypical pattern of resting frontal cortical asymmetry could be indicative of significant risk for the subsequent development of mood disorders. A meta-analysis from Thibodeau et al. (2006) observed similarly large effect sizes on this measure in studies examining clinically diagnosed subjects and those consisting of non-clinical subjects with higher anxiety or depression scores.

Another proposed EEG marker of anxiety disorder is right posterior alpha asymmetry (Bruder et al., 1997; Metzger et al., 2004), reflecting increased right hemispheric activity compared to left. Preliminary evidence has highlighted the utility of this metric in distinguishing EEG profiles that relate to anxiety versus depression (Nichske & Heller, 1998), although further research is warranted. It is important to note that given the higher mood disorder prevalence in females many of the reported EEG results have occurred in predominately female samples (Thibodeau et al., 2006). Therefore, it is inconclusive whether EEG traits relating to anxiety are different for males and females, although neither Moscovitch et al. (2011) nor Adolph and Margraf (2017) found an influence of gender using equal group sizes of males and females.

Current study

It is advantageous to extend this line of research to preadolescents (8–12 years), an age-range when anxiety and depression commonly onset (Simon et al., 2014). Many of the aforementioned training tasks (i.e. dual-n-back, EFMT) are geared toward adult levels of cognition; therefore, in this study we administered the gFocus training application (Smith, 2018) which is comparable to a go/no-go paradigm featuring emotional content. Thus, the primary aim of the current study was to assess the effects of computerized emotional IC training on anxiety, IC, and resting-state electroencephalography. The associations between these measures were also examined at baseline. Participants included three groups of non-clinical, preadolescent participants aged 8–12. The first group (A) was given 4 weeks of the emotional gFocus training (gFocus Resilience); the second (B) was given an identical 4-week gFocus training with neutral facial expressions (gFocus Classic), and the third (C) was

waitlisted for the training. Participants in groups A and B were given 16 sessions of training over the course of 4 weeks (4 days weekly). Four weeks has been deemed a sufficient length of computerized EF training to observe behavioral and neurological effects (Meyer et al., 2020).

Significant baseline associations between IC, anxiety, depression, and right frontal asymmetry were anticipated. Additionally, it was expected that the emotional training condition (gFocus Resilience) would show greater reductions in anxiety symptoms as well as an increased leftward shift in frontal alpha asymmetry (from pre-to-post) compared to the gFocus Classic and waitlist conditions. It was hypothesized that similar cognitive improvements (increased IC) would occur in both gFocus conditions relative to the control group.

Method

Participants

Participants at various schools near a South Florida university were sampled from fourth, fifth, and sixth-grade classrooms. Children who had been previously diagnosed with a developmental disorder, learning disability, or neurological condition according to parent reports were not included in the study. The participants consisted of 42, 8–12-year-olds (22 females, 20 males) with an average age of 10.12 ($SD = 1.25$). A full summary of the demographic information can be found in Table 1. Of the 42 participants, 32 completed the full 16 training sessions, and therefore this sub-sample was used for all analyses related to the effects of training. The G* Power software version 3.1 (Faul et al., 2017) was used a priori to determine appropriate sample sizes for detecting significant effects with a power level of .80 in the regression and MANCOVA analyses of interest with alpha set at .05 and the eta squared set at .14 using guidelines from (Bakeman, 2005). The recommended sample size to detect significant effects in the baseline regression models was at least 29 participants, while a minimum of 9 participants per training condition was advised for the multivariate tests evaluating training effects. All participants were treated in line with the ethical principles of the APA, and there was no risk of harm greater than normal classroom activities. All Participants received access to the gFocus software from IQ Mindware on their computer/tablet, which is normally a \$10–20 value.

Materials

Experimental condition: gFocus resilience (emotional)

The gFocus Resilience training program (Smith, 2018) presents participants with a continuous stream of blended stimuli, consisting of an emotional expression (male or female) projected in either an indoor or outdoor setting. These blended images are generated by altering pixel intensities through various weightings (default set at 50% face, 50% scene). In each training block, the participant is instructed to focus on one of four cues related to the images

Table 1. Demographic characteristics of the sample.

<i>N</i> = 42	Frequency/mean	Percent/ <i>SD</i>
participant's age	10.12 years	1.45
	8-year-olds: <i>N</i> = 6	14
	9-year-olds: <i>N</i> = 11	26
	10-year-olds: <i>N</i> = 12	29
	11-year-olds: <i>N</i> = 12	29
Participant's gender	Females: <i>N</i> = 22	52.38
	Male: <i>N</i> = 20	47.62
Mother's education	"Graduate school": <i>N</i> = 17	40
	"Bachelor's Degree": <i>N</i> = 15	36
	"Two Year Degree": <i>N</i> = 2	5
	"Some college": <i>N</i> = 8	19
	"High school degree": <i>N</i> = 1	2
Ethnicity	Caucasian: <i>N</i> = 25	60
	African American: <i>N</i> = 7	17
	Haitian/Caribbean: <i>N</i> = 4	10
	Latin-American: <i>N</i> = 3	7
	Asian-American: <i>N</i> = 3	7
Household yearly income	40,000–60,000: <i>N</i> = 5	5
	60,000–80,000: <i>N</i> = 12	29
	80,000–100,000: <i>N</i> = 8	19
	100,000–120,000: <i>N</i> = 4	10
	Over 120,000: <i>N</i> = 13	31
Age: control group	10.47 years	1.46
Age: emotional training	9.71 years	1.01
Age: neutral training	10.15 years	1.28
Gender: control group	Males: 7 Females: 8	46.7; 53.3
Gender: emotional training	Males: 7; Females: 7	50; 50
Gender: neutral training	Males: 6; Females: 7	46.15; 53.85
SCARED: control	<i>M</i> = 22.42, <i>SD</i> = 10.87	
SCARED: emotional	<i>M</i> = 25.30, <i>SD</i> = 7.07	
SCARED: neutral	<i>M</i> = 26.98, <i>SD</i> = 8.78	
SCARED parent: control	<i>M</i> = 17.19, <i>SD</i> = 10.88	
SCARED parent: emotional	<i>M</i> = 21.27, <i>SD</i> = 7.50	
SCARED parent: neutral	<i>M</i> = 21.98, <i>SD</i> = 11.51	
CDI: control group	<i>M</i> = 17.42, <i>SD</i> = 10.88	
CDI: emotional	<i>M</i> = 16.33, <i>SD</i> = 3.14	
CDI: parent: neutral	<i>M</i> = 16.82, <i>SD</i> = 8.78	

(male, female, indoor, outdoor) and in a "go/no-go" style procedure, they are to press the spacebar when the desired cue appears and withhold response when it does not appear. Each training session is approximately 12–15 min and includes eight blocks in total. The cued word is presented for 1 second on screen followed by a series of 50 trials (images) in each block, presented for 1 second with no inter-stimulus interval. In each block of training there are 75% go trials and 25% no-go trials; additionally, the training is adaptive in that the pixilation intensities change from trial to trial based on performance. When participants get a trial correct the proportion of task-relevant information in the stimulus is increased by 10% for the subsequent trial (until a ratio of 90% relevant to 10% irrelevant is reached). In contrast, when the user gets a trial incorrect the task-relevant information in the stimulus is reduced by 10% until a ratio of 10% relevant to 90% irrelevant is reached, intensifying the difficulty of the task. The task utilizes this closed feedback loop to positively reinforce participants with a simpler task for successfully maintaining sustained focus and inhibitory control, and to negatively reinforce participants for decreased performance by providing a more concealed stimulus and difficult task (thereby causing increased concentration). The emotional expressions serve as distractors to be inhibited for successful performance, and the task further demands IC because the subject must ignore distracting aspects of the image and withhold the dominant response

("go") on the no-go trials. Following each training block, the participant is shown their accuracy (percentage of correct trials out of 50) and if they improved from the previous block they are rewarded with the sounds of trumpets playing. The training paradigm is novel but theoretically based on the work of deBettencourt et al. (2015) who utilized a nearly identical training approach in college-aged participants for one session while recording fMRI. The researchers found that a 30-min training session in adults led to increased resting-state activation in frontal and ventral temporal cortices, the basal ganglia (striatum and globus pallidus), as well as the frontoparietal network. Additionally, significant improvements in behavioral performance were observed following the single session of training. The emotional version of the gFocus is hypothesized to train cognitive control in the presence of emotional content. Although this specific paradigm has only been tested in one study (deBettencourt et al., 2015), similar computerized training approaches have shown to successfully reduce attentional or cognitive impairments in a sample of adults with social anxiety disorder (Beard et al., 2011) as well a non-clinical adult sample scoring high on the Penn State Worry Questionnaire (Hayes et al., 2010). Notably, the gFocus app is child-friendly and 28 of the 32 participants responded that they were "very satisfied" with the training while four participants indicated that they were "somewhat satisfied". Answer choices to this follow-up question included very satisfied, somewhat satisfied, neutral, somewhat dissatisfied, and very dissatisfied.

Control conditions: gFocus neutral (traditional) and waitlist

The neutral gFocus training program also presents participants with a continuous stream of blended stimuli, consisting of a facial expression (male or female) projected in either an indoor or outdoor setting. This training is essentially identical to the emotional gFocus; however, in the traditional version all faces are neutral, eliciting no positive or negative affect. Additionally, a third group of participants consisted of waitlist controls to allow for further evaluation of the program efficacy and the importance of the emotional component. The waitlist control participants continued their typical school-related activities for the four-week time period and none indicated that they took part in any other form of cognitive or neurological training.

Inhibitory control assessments

Non-emotional or "cool" tasks were selected to assess IC in this study to allow a preliminary and general exploration of the anxiety-IC link in preadolescents. Hot or emotional IC tasks were not included in the test battery. In adult samples, the association between IC and anxiety has been demonstrated with non-emotional IC tasks (Edwards et al., 2017) and emotional ones (Yoon et al., 2017). The classic versions of the Stroop task and go/no-go task were administered using PsyToolkit (Stoet, 2017), and the flanker task through the Cognitivedun.net platform. All three tasks (Stroop,

flanker, go/no-go) consisted of 100 trials, preceded by 5 practice trials. Inhibitory control is considered a heterogeneous construct that can be divided into task-relevant and task-irrelevant abilities. Thus, the go/no-go was used to capture task-relevant inhibition (withholding response to red circles) and the flanker/Stroop tasks were used to encapsulate task-irrelevant inhibition (avoiding delay on incongruent trials).

In the *go/no-go*, stimuli were presented in a continuous stream and participants performed a binary decision on each stimulus. Participants were required to press the space bar in response to a green circle (go), whereas the other stimuli (red circle) required participants to withhold a response (no-go). Go events occurred 75% of the time and no-go events occurred 25% of the time. Each signal appeared for 2000 milliseconds and a valid response window was set to 0–2000 milliseconds post stimulus onset. Only accuracy (calculated as average no-go accuracy—average go accuracy) was used to gauge IC ability because reaction time is a proxy for processing speed in this task (Meule, 2017; Rai et al., 2017) given that it can only be measured for the congruent (go) trials.

A variation of the original *flanker task* (Eriksen & Eriksen, 1974) was administered through the cognitivefun.net platform and is described below: Participants needed to respond to a target stimulus, an arrow head (> or <), by pressing the corresponding arrow on the keyboard with their left index finger if the stimulus was pointing left (<) or with their right index finger if it was pointing right (>). The target stimulus was always the center stimulus in a display of five or more arrows. The flanker stimuli were either congruent with the target stimulus, (i.e. pointing the same way) or incongruent (i.e. pointing the opposite way) creating four trial types presented with equal likelihood (left congruent, right congruent, left incongruent, and right incongruent). The dependent measures for this task were interference accuracy (calculated as average incongruent accuracy—average congruent accuracy) and interference response time (average incongruent RT—average congruent RT).

In this version of the *Stroop test* (MacLeod, 1991) participants were instructed to indicate the color of ink that a word was written in while ignoring the connotation of the written word. Two different types of stimuli (congruent ink color, incongruent ink color) were presented along with four types of colors (blue, green, red, yellow). In this administration, 50% of the trials were incongruent (i.e. ink color does not match the word) and the other half were congruent (i.e. ink color matches the word). The dependent measures for this task were interference accuracy (average incongruent accuracy—average congruent accuracy) and interference response time, (average incongruent RT—average congruent RT).

Mood measures

The Screen for Child Anxiety Related Disorders (SCARED) is a 41-item inventory rated on a 3-point (0–2) scale (Birmaher et al., 1997). It comes in two versions; one asks questions to parents about their child and the other asks the

same questions to the child directly. The purpose of the instrument is to screen for signs of anxiety disorders and symptomology in children. For all participants, these forms were administered in-person by a trained member of the research team. For both the child and parent versions, the total score and the five factor scores (panic/somatic, generalized anxiety, separation anxiety, social phobia, school phobia) have demonstrated good internal consistency ($\alpha = .74-.93$), test-retest reliability (intraclass correlation coefficients = .70–.90), and moderate parent-child agreement ($r = .20-.47$, $p < .001$) (Birmaher et al., 1999). Higher scores are indicative of increased anxiety levels, and a total score of 25 or higher may indicate the presence of an anxiety disorder. In this sample, the parent-child agreement across items was moderately strong ($r = .216-.626$, $p < .001$); still, both metrics were used to analyze the construct of anxiety.

The Children's Depression Inventory (CDI)-short version is a 10-item screening tool used to rate the severity of symptoms related to depression disorder in children and adolescents (Helsel & Matson, 1984). The CDI-short version has shown strong internal consistency ($\alpha = .78$) and test-retest reliability in prior investigations ($\alpha = .82$) when compared to other instruments as well as the full version ($r = .86$, $p < .01$) (Allgaier et al., 2012). The CDI captures various elements of depressive disorder including negative mood, interpersonal problems, ineffectiveness, anhedonia, and negative self-esteem. Children rate themselves based on how they feel and think, with each statement being considered using a rating from 0 to 2. A cutoff score of 19 is recommended as a threshold for denoting clinical levels of depression.

The *Positive and Negative Affect Schedule for Children (PANAS-C)* was administered to assess a child's disposition for positive emotionality versus negative (Laurent et al., 1999). The PANAS-C is a 30-item self-report scale that measures both positive and negative affect, as indicated by ratings of single words on a 1–5 Likert scale, according to their personal relevance. Scores can range from 15 to 75 for each subcategory with higher scores signifying greater amounts of positive or negative affect reported (Laurent et al., 1999).

EEG measures

EEG was recorded for 6 min from pre frontal (FP1, FP2), anterior frontal (AF3, AF4), mid-frontal (F3, F4), lateral frontal (F7, F8) as well as parietal (P3, P4) leads which were referenced to the vertex (Cz). Prior to the positioning of the stretch lycra caps (manufactured by Electro-Cap, Inc.) head circumference was measured in centimeters, in order to fit the ideal cap size. Once positioned, omni prep gel and electrode gel were inserted into each site in order to gently abrade and provide good conductance. Impedances were checked and deemed acceptable if they were less than 5 K ohms. Eyes open data was collected and analyzed, and therefore the participant's mother and research team were instructed to remain quiet during the procedure. The EEG signals were amplified using Grass Model 12 Neurodata Acquisition System amplifiers with filters set at 1 Hz high

pass and 100 Hz low pass as well as a gain of 20,000, with 60 Hz notch filter on. Prior to data collection, the signal for each channel was calibrated using a 10 Hz sine wave. The analogue output from the amplifiers was then digitized at a sampling rate of 512 samples per second and digitized using 12-bit conversion (Analog Devices RTI-815 A/D board). The data were streamed to a computer screen and saved to a hard disk using data acquisition software (Snapstream, v. 3.21, HEM Data Corp. 1991).

Data reduction

Prior to transformation and segmentation into frequency bins, data were cleaned using the Artifact Subspace Reconstruction feature of the EEGLab toolbox in Matlab, which utilizes principle component analysis (PCA) to reject continuous data that is outside of five standard deviations of the cleanest point of data. The data were submitted to a discrete Fourier Transform using a 4-second Hanning Window with 50% overlap via the Matlab signal processing toolbox software (Mathworks) and then segmented into low alpha (6–9 Hz) and high alpha (8–12 Hz) frequency bands. For each participant, these analyses produced average power density values for the recorded regions of interest. Because power values are typically skewed and leptokurtotic, it is precedent to use the natural logarithms of the absolute power values for data analysis. This pattern was confirmed in this data set and therefore natural log transformed values were used for all power and asymmetry analyses. EEG asymmetry scores for all sites were calculated by subtracting the right hemispheric regional power from the left hemispheric regional power for the given site (i.e. $\text{Ln}(F4) - \text{Ln}(F3)$ for mid-frontal). Due to the inverse relation of alpha power to activity, negative asymmetry scores reflected greater right than left activity, and positive scores reflected greater left than right EEG activity.

Procedure

All participants came with their mothers for the initial baseline session where the pair was first consented, and then the participant's mother was given a demographic form capturing information related to age, ethnicity, and socio-economic status (SES). Next, three self-report instruments (SCARED-Child, CDI-short version, and PANAS-C) were completed by the child and one (SCARED-Parent) was completed by the mother. Following these questionnaires, the IC tests were administered in the same order for all participants as follows: go/no-go, flanker, and Stroop. Lastly, resting-state EEG was recorded for 6 min in total. At the end of the visit, participants were randomly assigned to one of three conditions (emotional gFocus, neutral gFocus, waitlist control), and the subsequent software was installed on their computer or tablet (both from IQ Mindware). A member of the research team assisted with the installation and set up a training profile and schedule that was consistent for each participant. They were instructed to complete the training (12–15 min) for a total of 16 sessions over the course of the

4 weeks, in-line with training recommendations from Apter (2012). Participants were instructed that they should complete the training during the weekdays and give their child the weekend to rest. Participants were asked to bring their computer back for the follow-up session so that the completion and duration of training could be confirmed by the research team. Following the 4-weeks of training, a follow-up appointment occurred for 32 of the 42 participants who successfully completed the training protocol. During this meeting the same measures from the baseline session were completed. Following the study, participants provided feedback and were allowed to keep the program.

Data analysis

First, a series of hierarchical linear regressions were conducted to analyze the associations between anxiety, depression, inhibitory control, and EEG asymmetry at the baseline time point. In these models, age and gender were controlled for in step 1, and the key predictor variable of interest was entered in step 2. For these analyses, the five dependent variables capturing inhibitory control were submitted to data reduction via principle component analysis. Next, to evaluate the effects of the IC intervention, three multivariate analyses of covariance (MANCOVAs) were generated separately for self-report mood measures (SCARED, CDI, PAS, NAS), inhibitory control accuracy (go/no-go, flanker, Stroop), and inhibitory control time (flanker, Stroop). The multivariate and univariate effects were analyzed and when appropriate, Tukey post-hoc tests were conducted to determine if group differences existed in change scores from pre-to-post. Age and gender were entered as covariates in these models but yielded no significant effects. Finally, two mixed-model ANCOVAs were generated to explore the effects of training condition and time on low and high alpha EEG measures, controlling for age/gender.

Results

Principle component analysis

For the regression models, the five dependent outcomes capturing inhibitory control (go/no-go accuracy, Stroop test accuracy, Stroop test time, flanker task accuracy, flanker task time) were submitted to a principle component analysis (PCA) with varimax rotation. The factor analysis yielded 2 factors with eigenvalues > 1 and factor loadings $> .5$, which together accounted for 83.13% of the variance in IC performance. The first factor accounted for 52.18% of the variance and included high loadings for go/no-go accuracy (.938), flanker accuracy (.903), and Stroop accuracy (.929); thus, this factor was labeled "IC accuracy". The second factor (labeled "IC Time") accounted for 30.92% of the variance and comprised high loadings for Stroop test time (.869) and flanker task time (.869). To simplify the analyses, factors IC accuracy and IC time were used for the baseline regressions.

Table 2. Partial correlations between anxiety, depression, and IC controlling for age and gender.

	Anxiety scale	Anxiety parent	Depression scale	IC accuracy	IC time
Anxiety scale					
Correlation		.478*	.514*	-.449*	.081
Significance (2-tailed)		.002	.001	.004	.620
Anxiety Parent					
Correlation	.478		.531**	-.606**	.307*
Significance (2-tailed)	.002		.000	.000	.050
Depression Scale					
Correlation	.514*	.531**		-.653**	-.150
Significance (2-tailed)	.001	.000		.000	.356
IC Accuracy					
Correlation	-.449*	-.606**	-.653**		-.064
Significance (2-tailed)	.004	.000	.000		.695
IC Time					
Correlation	.081	.307*	-.150	-.064	
Significance (2-tailed)	.620	.050	.356	.695	

**Correlation is significant at the 0.01 level (2-tailed).

*Correlation is significant at the 0.05 level (2-tailed).

Baseline associations of targeted variables

The first set of hierarchical regression models examined the predictiveness of IC accuracy and IC time on self-reported anxiety, parent-reported anxiety, and self-reported depression. In total, three models were conducted with each predictor (IC accuracy and IC time) and therefore, a Bonferroni-corrected alpha was used for significance criteria ($.05/3 = .017$.) The three emotionality variables were unsurprisingly all significantly correlated with one another, all p 's $< .001$, see Table 2. In each model, step 1 was not significant nor were the coefficients of age or gender at step 1 or 2. The model predicting self-reported anxiety from IC accuracy yielded significance at step 2, $F(3,38) = 3.49$, $p = .025$, $R^2 = .216$, $adjusted\ R^2 = .154$, $B = -4.13$, $t(41) = -3.10$, $p = .004$. Only the IC accuracy coefficient remained significant using the Bonferroni correction. Similarly, IC accuracy (at step 2) accounted for significant variation in parent-reported anxiety, $F(3,38) = 8.14$, $p < .001$, $R^2 = .391$, $adjusted\ R^2 = .343$, $B = -5.62$, $t(41) = -4.69$, $p < .001$, as well as self-reported depression $F(3,38) = 6.51$, $p < .001$, $R^2 = .45$, $adjusted\ R^2 = .410$, $B = -2.95$, $t(41) = -5.31$, $p < .001$. None of the overall models predicting anxiety or depression from IC time yielded significance, however in the model predicting parent-reported anxiety, IC time emerged as a significant coefficient, $B = 2.91$, $t(41) = 1.99$, $p = .05$, prior to the Bonferroni correction. Based on the correlations (Table 2) it appeared that anxiety was positively related with longer IC response latencies and IC inaccuracies. Depression was similarly related to increased inaccuracy, but decreased response latency. As a follow-up analysis to explore the influence of clinical levels of anxiety, participants were split into clinical-level anxiety (SCARED score ≥ 25 , $N = 20$) and non-clinical level anxiety (SCARED score < 25 , $N = 22$) groups. Independent samples t -tests showed that groups did not differ significantly on flanker reaction time or Stroop reaction time, but the high anxiety group displayed increased RT on the "IC time" factor (via the initial factor analysis) and increased flanker RT ($M = 358.20$, $SD = 80.43$) compared to the low anxiety group ($M = 351.41$, $SD = 74.57$). However, Stroop RT was slightly decreased in

the high anxiety group ($M = 403.70$, $SD = 71.21$) compared to the low anxiety group ($M = 408.59$, $SD = 146.49$).

Additionally, 30 hierarchical regression models were created to study the predictiveness of EEG low and high alpha asymmetry measures of interest (pre-frontal, anterior frontal, mid-frontal, lateral frontal, parietal) on the three emotionality variables. Once again, in each model, age and gender were entered at step 1 and the EEG asymmetry measure of interest at step 2. To control for the familywise error rate, a Bonferroni corrected p -value was used ($.05/15 = .003$) for the models using low alpha asymmetry measures and separately for the high alpha asymmetry models. None of the models showed significance at step 1 and neither age nor gender were significant coefficients at step 1 or 2. Only the model predicting self-reported anxiety from lateral frontal low alpha asymmetry yielded significance at step 2, $F(3,38) = 3.28$, $p = .03$, $R^2 = .205$, $adjusted\ R^2 = .143$, $B = -1.79$, $t(41) = -2.99$, $p = .005$. However, neither the model nor the coefficient remained significant with the Bonferroni correction. Regarding self-reported depression, none of the models were significant, however lateral frontal low alpha asymmetry was a significant predictor ($B = -.712$, $t(41) = 2.28$, $p = .028$) as was lateral frontal high alpha asymmetry ($B = -.682$, $t(41) = 2.22$, $p = .032$) before the Bonferroni corrections. In all instances, negative correlations suggested that increases in anxiety were related to increased right frontal activity compared to left (see Tables 3 and 4).

Factors influencing the likelihood to complete training

Given that 10 of the original 42 participants did not complete the full training program, a sub-sample of 32 was used to evaluate the effects of training on the outcomes of interest. Independent samples t -tests were conducted to determine if those who completed the study had any significant differences from those who did not complete the study on the following measures at baseline: self-reported anxiety, self-reported depression, parent-reported anxiety, go/no-go reaction time and accuracy, flanker reaction time and accuracy, and Stroop reaction time and accuracy. The group that

Table 3. Partial correlations between anxiety, depression, and low alpha asymmetry controlling for age and gender.

	Anxiety child	Depression child	Anxiety parent	FPA	AFA	MFA	LFA	PA
Child anxiety								
Correlation		.514*	.478*	.010	-.247	-.256	-.378*	.098
p-Value		.001	.002	.950	.124	.111	.016	.549
Child Depression								
Corr.	.514*		.531**	-.284	-.266	-.277	-.348*	-.218
p-Value	.001		.000	.076	.097	.084	.028	.177
Anxiety Parent								
Corr.	.478*	.531**		-.073	-.046	-.116	-.256	.077
p-Value	.002	.000		.655	.780	.477	.111	.635
Pre-Frontal								
Corr.	.010	.284	.073		.074	.397*	.052	.072
p-Value	.950	.076	.655		.650	.011	.752	.661
Anterior frontal								
Corr.	.247	.266	.046	.074		.226	.335*	.052
p-Value	.124	.097	.780	.650		.160	.034	.748
Mid-frontal								
Corr.	.256	.277	.116	.397	.226		.189	.322*
p-Value	.111	.084	.477	.011	.160		.242	.043
Lateral frontal								
Corr.	.378*	.348*	.256	.052	.335*	.189		.145
p-Value	.016	.028	.111	.752	.034	.242		.374
Parietal								
Corr.	.098	.218	.077	.072	.052	.322*	.145	
p-Value	.549	.177	.635	.661	.748	.043	.374	

**Correlation is significant at the 0.01 level (2-tailed).

*Correlation is significant at the 0.05 level (2-tailed).

Table 4. Partial correlations between anxiety, depression, and high alpha asymmetry controlling for age and gender.

	Anxiety Child	Depression Child	Anxiety Parent	FPA	AFA	MFA	LFA	PA
Child anxiety								
Correlation		.514*	.478*	-.115	-.298	-.292	-.437*	.012
p-Value		.001	.002	.479	.061	.068	.005	.942
Child depression								
Corr.	.514**		.531**	.207	-.264	-.259	-.339*	.226
p-Value	.001		.000	.200	.100	.107	.032	.070
Anxiety parent								
Corr.	.478*	.531**		-.035	-.060	-.019	-.151	.045
p-Value	.002	.000		.833	.713	.907	.352	.781
Pre-frontal								
Corr.	-.115	.207	-.035		.198	.074	.079	.306
p-Value	.479	.200	.833		.220	.652	.628	.055
Anterior frontal								
Corr.	.298	.264	.060	.198		.429*	.382*	.183
p-Value	.061	.100	.713	.220		.006	.015	.257
Mid-frontal								
Corr.	.292	.259	.019	.074	.429*		.232	.469*
p-Value	.068	.107	.907	.652	.006		.150	.002
Lateral frontal								
Corr.	.437*	.339*	.151	.079	.382*	.232		.161
p-Value	.005	.032	.352	.628	.015	.150		.322
Parietal								
Corr.	.012	.326*	.045	.306	.183	.469*	.161	
p-Value	.942	.040	.781	.055	.257	.002	.322	

**Correlation is significant at the 0.01 level (2-tailed).

*Correlation is significant at the 0.05 level (2-tailed).

did not complete the training ($M = 695.48$, $SD = 12.14$) had significantly increased go/no-go reaction time (milliseconds) at baseline compared to the group that completed the full training regimen ($M = 570.98$, $SD = 101.75$), $t(33.65) = -6.77$, $p < .001$. However, accuracy on the go/no-go was comparable between the group completing training ($M = 93.31$, $SD = 5.17$) and the group that did not ($M = 9.35$, $SD = 2.61$). None of the other measures of interest were significantly different between groups.

A series of chi-squared tests were also conducted on the two groups relating to demographic characteristics including age, gender, maternal education, and family yearly income.

Only the test for maternal education revealed significant differences between groups, $\chi^2(4, n = 42) = 15.20$, $p = .004$, signifying that maternal education predicted likelihood to complete training. In the group that completed training, one mother completed high school, 11 completed some college, one completed a two-year degree, two completed a bachelor's degree, and 17 completed graduate school. Alternatively, the group that did not complete training had six mothers that completed high school, four that completed some college, one that completed a two-year degree, five with a bachelor's degree, and zero that completed graduate school.

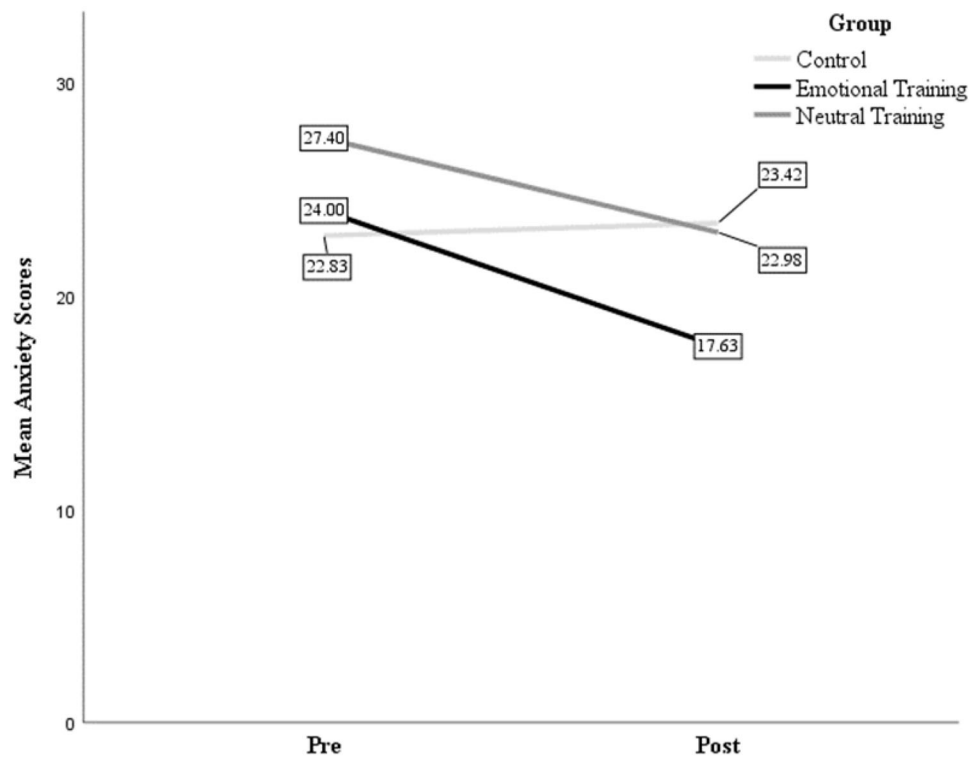


Figure 1. Group $\times$ Time Interaction (Child-Reported Anxiety). The 3×2 significant interaction effect of group and time on anxiety is visualized above. The emotional training condition ($M_{\text{change}} = -6.37$, $SE = 2.50$) showed a significantly larger reduction than the wait list control ($M_{\text{change}} = .58$, $SE = 2.61$), $p = .025$, but did not differ from the neutral training condition ($M_{\text{change}} = 4.42$, $SE = 2.50$), $p = .44$.

Training effects on self-reported anxiety and depression

To analyze the effects of training condition on self-reported mood measures from pre-to-post, a 2×3 multivariate analysis of covariance (MANCOVA) was conducted with training condition as the between-subjects variable (emotional training, neutral training, waitlist control) and time (pre/post) as the within-subjects variable. The dependent measures were self-reported anxiety, depression, positive affect, and negative affect scores. Parent-reported anxiety was not included in this analysis because the other four measures were rated by the child. As hypothesized, there was a significant two-way interaction effect of training condition $\times$ time on mood measures, $F(8, 48) = 6.79$, $p < .001$; $Wilk's \Lambda = 0.22$, $partial \eta^2 = .53$. The univariate tests highlighted significant effects of training condition $\times$ time on self-reported anxiety, $F(2, 27) = 5.22$, $p = .02$; $partial \eta^2 = .28$, self-reported depression, $F(2, 27) = 20.02$, $p < .001$, $partial \eta^2 = .57$, as well as self-reported negative affect $F(2, 27) = 5.91$, $p = .007$, $partial \eta^2 = .31$. There was however no effect on self-reported positive affect ($p = .515$). Tukey post-hoc tests were generated to examine which training conditions were most effective in altering scores from pre-to-post. Regarding anxiety, the emotional training condition ($M_{\text{change}} = -6.37$, $SE = 2.50$) showed a significantly larger reduction than the waitlist control ($M_{\text{change}} = -.58$, $SE = 2.61$), $p = .025$, but not the neutral training condition ($M_{\text{change}} = -4.42$, $SE = 2.50$), $p = .44$. The neutral training was not significantly different from the wait list control condition, $p = .13$ (see Figure 1).

For the self-reported depression outcome, Tukey post-hoc tests on the change scores showed that the neutral training condition ($M_{\text{change}} = -3.12$, $SE = .66$) had the largest reduction in depression which was significantly different from the emotional training condition ($M_{\text{change}} = -.96$, $SE = .66$), $p = .011$ as well as the waitlist group ($M_{\text{change}} = 1.08$, $SE = .69$), $p < .001$. The emotional training condition also led to significant reductions compared to the wait list control, $p = .012$ (see Figure 2). Tukey post-hoc tests on the measure of negative effect showed that the emotional training condition ($M_{\text{change}} = -4.38$, $SE = 1.52$) as well as the neutral training condition ($M_{\text{change}} = -3.82$, $SE = 1.52$) resulted in a significantly larger reduction in scores than the waitlist control ($M_{\text{change}} = .50$, $SE = 1.59$), $p = .035$ and $p = .045$, respectively (see Figure 3).

Training effects on inhibitory control

Similarly, A 2×3 MANCOVA was conducted to analyze the effects of the intervention on inhibitory control accuracy measures from pre-to-post. The between-subjects variable was group, the within-subjects was time (pre/post), and the dependent measures were go/no-go, flanker, and Stroop accuracy. The multivariate interaction effect of training condition $\times$ time on IC measures was significant, $F(6, 50) = 2.79$, $p = .019$; $Wilk's \Lambda = 0.58$, $partial \eta^2 = .23$. The univariate tests showed significant effects of training condition $\times$ time on go/no-go accuracy, $F(2, 27) = 6.01$, $p = .007$; $partial \eta^2 = .31$ and flanker accuracy, $F(2, 27) = 5.28$,

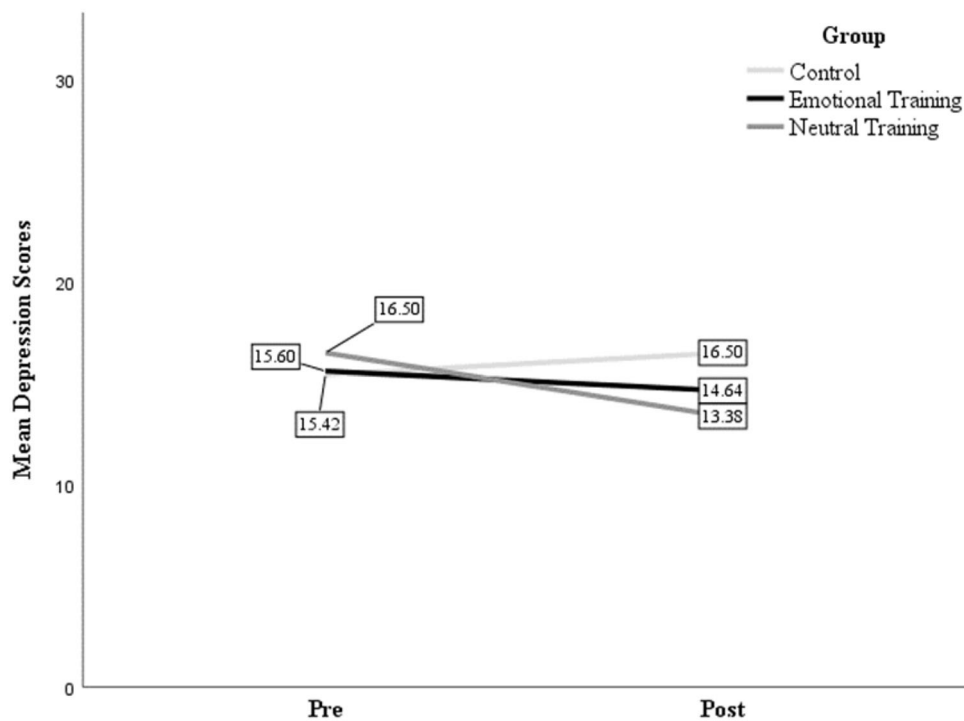


Figure 2. Group $\times$ Time Interaction (Child-Reported Depression). The 3×2 significant interaction effect of group and time on depression is visualized above. The neutral training condition ($Mchange = -3.12$, $SE = .66$) was related to the largest reduction in depression which was significantly different from the emotional training condition ($Mchange = -.96$, $SE = .66$), $p = .011$ as well as the control group ($Mchange = 1.08$, $SE = .69$), $p < .001$. The emotional training condition also led to significant reductions compared to the waitlisted control, $p = .012$.

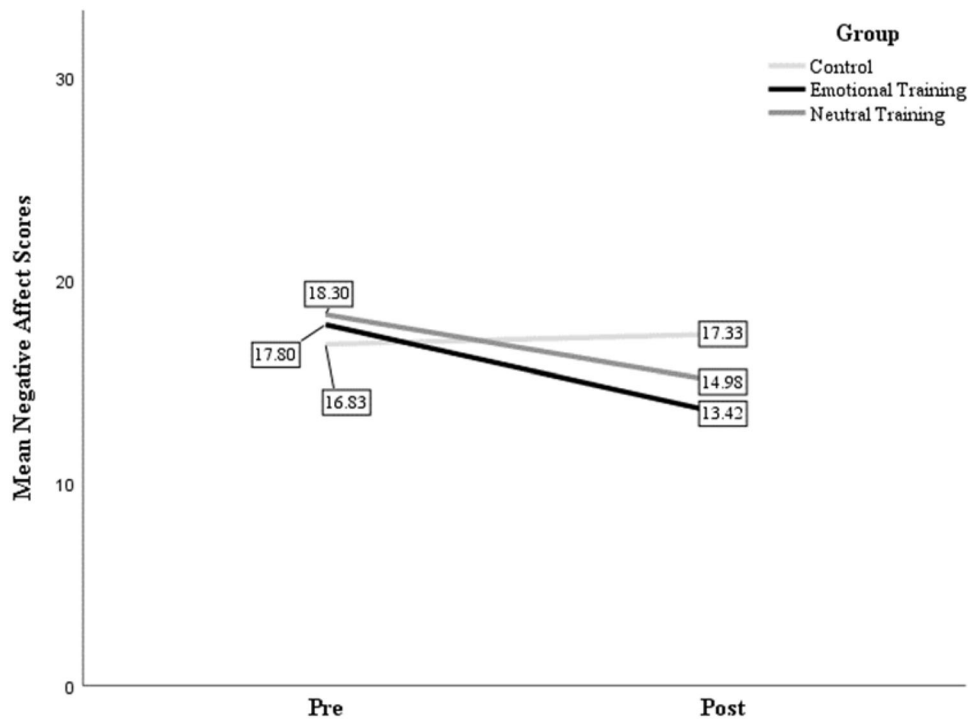


Figure 3. Group $\times$ Time Interaction (Child-Reported Negative Affect). The 3×2 interaction effect of group and time on negative affect is visualized above. The emotional training condition ($Mchange = -4.38$, $SE = 1.52$) as well as the neutral training condition ($Mchange = -3.82$, $SE = 1.52$) had significantly larger score reductions than the waitlist control ($Mchange = .50$, $SE = 1.59$), p 's $< .05$.

$p = .012$, $partial \eta^2 = .28$. The effect was marginally significant for Stroop accuracy, $F(2, 27) = 3.08$, $p = .062$, $partial \eta^2 = .19$. Tukey follow-up tests on the go/no-go accuracy change-scores showed that the neutral training condition ($Mchange = 6.40$, $SE = 1.75$) yielded significantly greater

improvements in accuracy than the control ($Mchange = .16$, $SE = 1.82$), $p = .004$. No other differences were uncovered with the emotional training condition ($Mchange = 2.44$, $SE = 1.75$). Regarding the flanker test, neutral training ($Mchange = 3.91$, $SE = 1.45$) resulted in significantly greater

improvements in accuracy compared to the waitlist control ($Mchange = -.011$, $SE = 1.75$), $p = .03$). as well as the emotional training condition ($Mchange = -.66$, $SE = 1.52$, $p = .014$). Finally, regarding the Stroop test, no changes occurring between the groups were significant. These results are found in Table 5.

An identical 2×3 MANCOVA was conducted to analyze the effects of treatment on IC time (duration). As hypothesized, the interaction effect of group $\times$ time on IC time per trial was significant, $F(6, 50) = 3.35$, $p = .007$; $Wilk's \Lambda = .51$, $partial \eta^2 = .29$. The univariate tests showed that the effects of group $\times$ time were only significant on flanker trial RT, $F(2, 29) = 5.11$, $p = .013$, $partial \eta^2 = .27$. A Tukey follow-up test on the change scores showed that both the emotional training condition ($Mchange = -19.40$, $SE = 5.40$) and the neutral training condition ($Mchange = -17.70$, $SE = 4.48$) had significant flanker task time reductions compared to the control group ($Mchange = -4.50$, $SE = 3.98$), $p = .017$ and $p = .031$ respectively. In summary, flanker task and go/no-go accuracy were improved in both the emotional and neutral training groups relative to control, and flanker reaction time was reduced.

Training effects on resting-state neurophysiology

To analyze the effects of training on resting-state low alpha EEG, a $3 \times 2 \times 5 \times 2$ mixed-model ANCOVA was conducted with between-subjects factor group (emotional training, neutral training, waitlist control), and within-subjects factors time (pre, post), region (pre-frontal, anterior frontal, mid-frontal, lateral frontal, parietal) and hemisphere (left, right). Age and gender were also entered as covariates in these analyses. Only the main effect of time was significant $F(1, 27) = 6.71$, $p = .015$, $partial \eta^2 = .20$; additionally, the only significant two-way interaction effect was time $\times$ age, $F(1, 27) = 6.80$, $p = .015$, $partial \eta^2 = .20$. The four-way interaction effect of group, time, region, and hemisphere was non-significant, $F(8, 108) = 1.04$, $p = .414$, $partial \eta^2 = .07$, however the three-way interaction effect group $\times$ time $\times$ hemisphere was of particular interest for determining whether the training induced changes in regional asymmetry. This effect was marginally significant, $F(8, 116) = 3.62$, $p = .061$, $partial \eta^2 = .20$. Exploratory one-way ANOVAs were conducted to examine group differences on the asymmetry change scores from pre-to-post in the regions of interest (pre-frontal, anterior frontal, mid-frontal, lateral frontal, parietal). Significant differences from pre-to-post were only detected in the lateral frontal region ($F(2, 31) = 3.65$, $p = .03$). This was quantified by a leftward shift in activity in the emotional training condition ($Mchange = .956$, $SE = 1.61$) compared to the wait list control ($Mchange = -1.15$, $SE = 2.09$), $p = .039$ via Tukey tests. A replicated ANCOVA model with high alpha values was created but did not yield any significant four-way or three-way interaction effects, all $p's > .05$. However, the main effect of time and interaction of time and age were again significant.

Table 5. Inhibitory control changes from pre-to-post.

Group	GNG accuracy pre	GNG accuracy post	GNG time pre	GNG time post	Flanker accuracy pre	Flanker accuracy post	Flanker time pre	Flanker time post	Stroop accuracy pre	Stroop accuracy post	Stroop time pre	Stroop time post
Control												
N	15	12	15	12	15	12	15	12	15	12	15	12
M	-5.42	-4.33	577.35	538.79	-5.72	-5.50	347.60	336.75	-9.48	-8.00	436.73	440.08
SD	1.18	1.14	116.16	117.11	1.72	1.56	63.63	59.81	2.56	2.55	163.40	192.49
Emotion training												
N	14	10	14	10	14	10	14	10	14	10	14	10
M	-6.25	-3.96	591.65	543.59	-7.01	-7.07	362.14	347.50	-8.14	-5.89	393.78	371.20
SD	1.11	1.37	97.51	75.92	1.74	1.81	86.63	265.81	2.33	2.83	82.04	83.04
Neutral training												
N	13	10	13	10	13	10	13	10	13	10	13	10
M	-8.50	-3.20	637.15	552.28	-8.80	-5.47	354.69	343.60	12.02	7.48	384.54	362.00
SD	2.33	1.42	91.84	53.36	1.50	1.32	83.86	85.96	3.10	2.77	72.86	90.27

Discussion

Here, the aim was to expand upon research investigating cognitive and neurological mechanisms involved in childhood anxiety and depression and to evaluate the efficacy of a cognitive training regimen for mitigating negative emotionality symptoms. Primarily, we explored the associations between anxiety, depression, and inhibitory control abilities using three tasks (go/no-go, flanker, and Stroop) and found that IC accuracy was significantly and negatively related to anxiety as well as depression. IC reaction time was not significantly related to these emotional measures, although there was a trend for increased reaction time with increased anxiety symptomology. Additionally, lateral frontal alpha asymmetry was predictive of anxiety and depression at baseline, and correlations confirmed the common finding of greater right frontal asymmetry in relation to mood symptoms. Another primary aim was to evaluate the effects of a 4-week, 16 session computerized IC training program. In summary, the emotional and neutral training conditions led to significant reductions in anxiety, depression, and negative affect relative to the waitlist group, with the emotional training condition showing the largest reductions in anxiety and negative affect. These two conditions showed comparable improvements in IC accuracy relative to the waitlist, with greater increases observed in the neutral training condition. Significant differences in change scores were seldom apparent between the emotional and neutral training conditions across all outcomes. Notably, In the emotional training condition there was a significant leftward shift in lateral frontal alpha asymmetry relative to the wait list control.

Baseline associations between negative emotionality and IC accuracy

The finding of significant associations between mood measures, IC measures, and neurophysiology in a preadolescent sample is novel and warrants further discussion. IC accuracy (comprised of go/no-go, Stroop, and flanker) predicted significant variation in self-reported anxiety and depression, as well as parent-reported anxiety, whereas IC time (comprised of flanker and Stroop time) did not. In a comparable investigation consisting of adults with anxiety aged 30 to 45 years, Hallion et al. (2017) found that Stroop accuracy was significantly depleted, and reaction time increased, however go/no-go performance was not altered. In the present study, the mixed findings related to IC reaction time is contradictory to Hallion et al. (2017) as well as Eysenck et al. (2007), who found significantly increased antisaccade reaction time in an adult sample with pathological anxiety. Still, there has been debate on whether RT will present as increased or decreased in relation to anxiety. Edwards et al. (2017) found that IC RT was decreased as a function of anxiety; a finding that may reflect on challenges with suppressing an initial bottom-up response, in order to answer the trial correctly. Mueller et al. (2012) replicated this finding in a sample of

adolescents on an emotional antisaccade task. The finding in this current study alludes to the possibility that some children coping with negative emotionality will show stunted or delayed reaction time due to concentration difficulties stemming from rumination or worry, while others may show impulsive or increased reactions due to difficulties suppressing their initial response. It will be worthwhile for future research to examine the IC-anxiety relationship as a function of different types of anxiety (general versus panic), as well as effects of the co-presence of externalizing traits. Self-reported depression was also strongly related to IC accuracy, which further demonstrates the comorbidity of anxiety and depressive symptoms and suggests that IC is a cognitive mechanism involved in both conditions.

It should also be noted that traditional or “cool” tasks were used to capture the general construct of inhibitory control in this study. Some prior studies have utilized traditional EF tasks to study associations between IC and anxiety, while others (De Voogd et al., 2016) have used “hot” or emotional tasks, often producing comparable results. Here, traditional tasks were used as a preliminary approach to testing the IC-anxiety link in a novel age-range. Hot or emotional EF tasks were not administered in this initial study; however, we recommend that future studies seek to determine if the link between anxiety and IC is stronger when using emotional IC tasks. It would have also been compelling to assess whether the emotional gFocus facilitated greater improvements in emotional IC compared to traditional IC.

Baseline associations between negative emotionality and resting-state EEG

At baseline, the only EEG measures that predicted child-reported anxiety were right lateral frontal low alpha and high alpha asymmetry scores. No significant regression models or correlations emerged examining alpha asymmetry in relation to parent-reported anxiety, however right lateral frontal alpha asymmetry was similarly predictive of increased child-reported depression. The finding of a relation between right lateral frontal asymmetry and negative emotionality is in line with numerous prior investigations on this topic (Thibodeau et al., 2006). Increased right activity has been cited as a neural marker of avoidant, withdrawn, and anxious behavior, whereas increased left signifies a greater likelihood for motivated, approach behavior (Coan & Allen, 2004). It is somewhat surprising that parietal asymmetry was not also associated with anxiety (increased right activation) as this measure has been previously implicated using adult samples (Bruder et al., 2017). Still, it should be noted that this was a non-clinical sample, creating less natural fluctuation in EEG power values which may have partially influenced some of the null findings. There is also preliminary evidence to show that both theta and beta power and asymmetry values are reduced as a function of childhood anxiety (Demerdzieva, 2011). Therefore, it may be

worthwhile to explore additional EEG measures across higher and lower activity frequencies in future investigations. Child-reported depression was similarly related with frontal asymmetry in this sample, further demonstrating the difficulty of teasing apart neural correlates of anxiety and depression. Bruder et al. (2017) has shown that parietal asymmetry may present differently in the two conditions, however this did not occur here. A meaningful direction for future work is to examine the possibility that brain activity during the processing of IC tasks serves as a mediator between anxiety levels and performance.

The effects of gFocus IC training

In line with our hypotheses, the intervention facilitated improvements in the targeted skills, as go/no-go and flanker accuracy were significantly improved, and Stroop accuracy was moderately improved. Flanker reaction time was also decreased from pre-to-post. It is possible that IC reaction time was altered to a lesser extent because the gFocus task does not change speeds or require a user to respond quicker as levels increase (Smith, 2018). A user is rewarded for increasing their accuracy, which may have led to a slower-processing strategy relying on top-down neurocircuitry as opposed to quicker, bottom-up strategies. In this sample, mixed findings occurred for reaction time in relation to anxiety and depression at baseline, potentially contributing to the lack of change observed over time.

Notably, statistically significant differences between the emotional and neutral training conditions were scarcely apparent, which was at odds with our hypothesis stating that emotional training would facilitate significantly greater reductions in mood symptoms compared to the other two conditions. This finding was evidence that training with emotional content is not essential for improving EF skills or emotional functioning as some have suggested in the past (Iacoviello et al., 2014). The lack of differential effects between these two conditions could be attributed to three possibilities: (1) That the improvements in inhibitory control which occurred in both conditions was enough to reduce anxiety, regardless of whether emotional expressions were included; (2) that participants in the neutral condition still experienced “emotional EF training” to some extent given that their task featured neutral facial stimuli and not shapes or other non-emotional content; (3) that the group sample sizes ($N=10$) were not sufficient to capture statistically significant differences in the outcomes. The emotional training condition did lead to greater reductions in both anxiety and negative affect compared to the neutral training. With a larger sample size, it is possible that these effects would have trended toward significance.

As an additional means for training evaluation, resting-state EEG low and high alpha asymmetry were examined in regions found to be related to anxiety and/or EF in prior studies. Group differences in asymmetry change scores only occurred in the lateral frontal region where there was a leftward shift in activity in the emotional

training condition compared to the waitlist control. Although the training did not facilitate any additional significant neurophysiological changes suggesting reduced risk of anxiety, the leftward shift of frontal alpha asymmetry was highly encouraging. It is possible that the 16 session protocol was not sufficient to change other neurophysiological characteristics, although this training amount was in line with past studies (Bomyea et al., 2015). Still, it will be necessary to explore lengthier training procedures with additional assessment time points for future studies.

Limitations and future directions of research

There were limitations of this investigation as well as intriguing opportunities for future research. First of all, although the sample size was in-line with some prior investigations (and deemed appropriate in G-power analyses) an increased sample size would have led to increased power for detecting differences between the three groups, particularly on EEG results which are often subtler than self-reported measures. Additionally, a lengthier protocol of training would have allowed long-term effects to be studied in greater detail, as well as the presence of a ceiling effect. Another limitation to the study was that by using resting-state EEG measures, we were not able to directly study how brain activation patterns during a given IC task relate to negative emotionality. Using event-related potentials would have allowed us to better identify the brain circuitry being trained in the gFocus task as well as the necessary circuitry for completing the IC assessments. Future studies can also strengthen this research by embedding an emotional IC task into the test battery, and by presenting the tasks in a varied order. Here, the three IC tasks were completed by all participants in a fixed order and thus, IC scores may have been confounded by ordering effects in this study. Finally, there are fascinating opportunities to examine the role of pathological anxiety and depression to a larger extent or the possibility that IC impairments are only related to subtypes of anxiety such as panic. Despite the promising results here, approximately 25% of the initial participants were not able to complete four weeks of short-duration training remotely. Devising programs that are more child-friendly or gamified will be essential for maximizing training accessibility and minimizing attrition.

In conclusion, this study provided preliminary evidence for the effectiveness of computerized IC training in mitigating negative emotionality symptoms in preadolescent children. It was also encouraging that frontal alpha asymmetry shifted leftward in participants completing an emotional version of the training. At the baseline time point, there was further evidence to support the link between inhibitory control dysfunction and anxiety/depression. Decreased IC performance predicted higher levels of anxiety and depression, signifying that inhibitory impairments could be a risk factor for the development of these conditions in children. In the current social climate of the

world, internalizing conditions like anxiety and depression are becoming increasingly common in children and adolescents. Meanwhile, the availability and accessibility of computer and tablet technology has also rapidly increased. Providing computerized cognitive training programs to children can be a highly beneficial use of this technology for improving not only academic performance, but as seen here, psychological and emotional functioning during a challenging time-period of development.

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